

Zarin Tasnim Roichi

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SUMMARY

Computer Science graduate with research experience in machine learning, deep learning, and intelligent systems. Passionate about conducting meaningful research, developing practical solutions, and exploring new ideas. Seeking funded PhD opportunities where I can apply my existing skills while developing expertise in new research areas.

EDUCATION

2021 – Dec 2025	B.Sc. in Computer Science & Engineering (Completed) Ahsanullah University of Science & Technology (AUST) CGPA: 3.405 / 4.00
2019	Higher Secondary Certificate (H.S.C.) Holy Cross College, Dhaka GPA: 5.00 / 5.00
2017	Secondary School Certificate (S.S.C.) A.V.J.M Govt. Girls High School GPA: 5.00 / 5.00

SKILLS

Machine Learning & AI	Deep Learning, Supervised Learning, NLP (Bengali), Computer Vision, Multimodal Learning (Text + Image), Transformer Models, Large Language Models (LLMs), LoRA/PEFT Fine-tuning, Ensemble Learning (Stacking, Voting), Model Fusion (Late & Intermediate), Prompt Engineering, Model Evaluation (Precision/Recall/F1)
Frameworks & Libraries	TensorFlow (Keras) , PyTorch , Hugging Face Transformers, Scikit-learn, NumPy, Pandas
Programming	Python (Primary) , Java, C, C++, JavaScript, PHP
Deployment & Backend	ML Model Integration (Python APIs), MySQL, MSSQL
Tools	Git/GitHub, VS Code, Visual Studio, MATLAB, Android Studio

THESIS & PROJECTS

AI / Machine Learning Projects

Multimodal Fake News Detection (Thesis Project)

Multimodal (text + image + vision-language) fake news classification system.

- Evaluated seven BanglaBERT variants and built an MLP-based stacking ensemble for text classification.
- Trained diverse vision backbones (MobileNet, ResNet, Swin-T/B/L, ConvNeXt, ViT, DeiT, DiT) and constructed image-only ensembles (Best-3 / All-6).
- Integrated CLIP and BLIP for cross-modal embedding alignment.
- Implemented late fusion (α -weighted probability sweep) and intermediate feature-level fusion.
- Achieved up to **84.11% accuracy** using multimodal ensemble fusion.

Manuscript in preparation.

Bengali Fake Review Detection

Transformer- and LLM-based Bengali fake-review classification pipeline.

- Fine-tuned BanglaBERT variants, XLM-R, mBERT, BanglaT5, and Gemma-2B using **PyTorch** and Hugging Face.
- Implemented supervised LLM classifiers across 1B–7B scales, including Bangla-LLaMA, TituLLM, and Qwen, using classification heads and LoRA/QLoRA-based PEFT.
- Conducted causal-LM fine-tuning, zero-shot/few-shot prompting, prompt-sensitivity analysis, and zero-shot evaluation of provider-hosted LLM APIs.
- Built ensemble models (majority, soft, weighted voting, and stacking-MLP) using softmax probability fusion.
- Applied training-only augmentation and class rebalancing while preserving the original validation and test distributions; achieved up to **99.12% weighted F1-score**.

Manuscript in preparation.

Sentiment Analysis of Bengali Movie Reviews

Transformer-based Bengali sentiment classification.

- Fine-tuned BanglaBERT, mBERT, and XLM-R using PyTorch.
- Applied LoRA/PEFT-based parameter-efficient fine-tuning.
- Evaluated using accuracy, precision, recall, and F1-score (**97.16% accuracy**).

Software Development Projects

Employee Time & Attendance Tracker

Tech Stack: PHP, MySQL

- Developed a role-based attendance management system with secure authentication and relational SQL backend.

TimeVista

Tech Stack: ASP.NET, C#, MSSQL

- Built a structured attendance platform with optimized database queries and modular backend design.

AUST Connect

Tech Stack: Java, Firebase

- Implemented authentication, real-time data synchronization, and cloud-based backend integration.

Bus Ticket Management System

Tech Stack: Java, JavaFX, MySQL

- Designed and developed a booking system with MySQL-backed transaction handling.

Hardware & IoT Project

Offline Voice-Controlled Home Automation System

- Developed an Android-based offline speech-recognition system integrated with Bluetooth-controlled hardware modules.
- Designed rule-based command mapping and RFID-based access control mechanism.

Manuscript in preparation.

LANGUAGE PROFICIENCY

IELTS Academic: 7.0
